PHYSICAL CHEMISTRY CUMULATIVE EXAM

September 18, 1993

In Statistical Thermodynamics:

1.) What are the characteristic properties of

a) Fermions (Fermi-Dirac particles)
b) Bosons (Bose-Einstein particles)
c) Boltzons (Maxwell-Boltzmann particles)

In each case name some really existing examples.

2.) Assume that the number of wave functions of a system with energy ε_j is g_j and that we have n_j molecules of energy ε_j to be found in one or another of the g_j possible degenerate states. A particular distribution, (n_j), of molecules among the various energy levels is described by the number of molecules in each energy level:

$$(n_j) = (n_0, n_1, n_2, n_3, \ldots)$$

Each different way of assigning n_j indistinguishable particles among g_j logically and physically distinct but indistinguishable wave functions is a microscopically separate state, or **microstate**. All of these individual microstates produce the same macroscopically observable state or **macrostate**. These macrostates are just the distributions, (n_j).The fundamental principle of statistical mechanics is that, energy considerations aside, one microstate is just as probable as any other. Therefore $P(n_j)$, the probability of occurence of any given macrostate is proportional to $W(n_j)$, the number of possible microstates leading to it (the number of microstates per macrostate).

a)Prove the following formula for Fermions, giving the total number of microstates per macrostate, or the total number of ways of obtaining a distribution of many levels :

$$W_{(n_j)} = \prod_j \frac{g_j!}{n_j!(g_j - n_j)!}$$

b)Derive the corresponding formula for W(nj) for Bosons.